3D Printer → Chess Robot

Autonomous Chess-Playing Robot

Repurposing a Creality Ender 3 Pro with Computer Vision, AI & Electromagnetic Actuation

B5#RO Case Study | 2025/2026

Arne Jacobs · Sai Neha Narendra Babu · Daniel MGgurk · Vivek Devarapalli · Christian Lazaro

The Problem

Why Chess?

Chess demands precise X-Y-Z Cartesian motion, real-time decision-making, and complex vision-to-motion mapping, allowing for a perfect testbed for integrated robotics.

The Mission

Build a fully autonomous chess-playing robot by repurposing a Creality Ender 3 Pro using digital AI and machine vision.

Design Constraints

REQUIREMENTS

Fully Autonomous

Zero human intervention during gameplay after the first move.

Vision-Based

Camera detects board-state changes after every human move.

Budget Limited

£50 total budget, hardware components supplied from campus labs.

Safe Operation

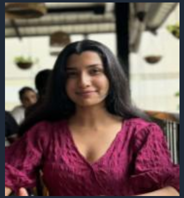
Low-speed motion; office-safe.

The Team

Arne Jacobs

Systems Architect & Integration Lead

High-level design
Hardware procurement
Soft-to-hardware integration



Sai Neha Narendra Babu

Research Analyst & Literature Lead

Benchmarking of existing state-of-the-art systems

Daniel McGurk

CAD Designer

Handled the 3D-printing workflow for the chess pieces

Vivek Devarapalli

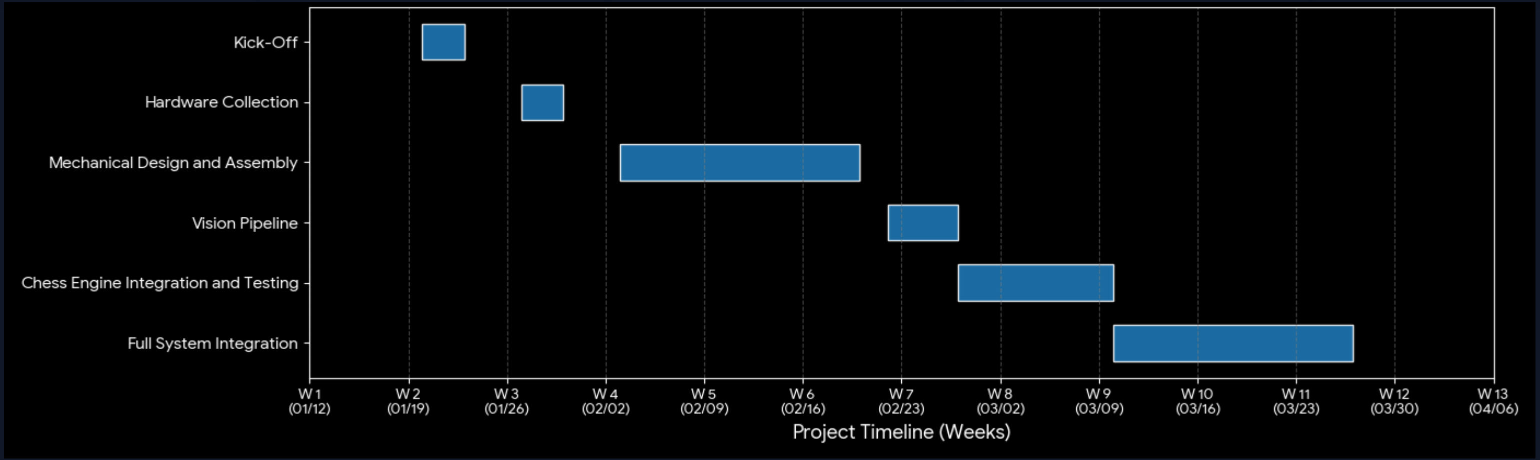
Embedded Systems Consultant

Technical advisory on firmware optimization

Christian Lazaro

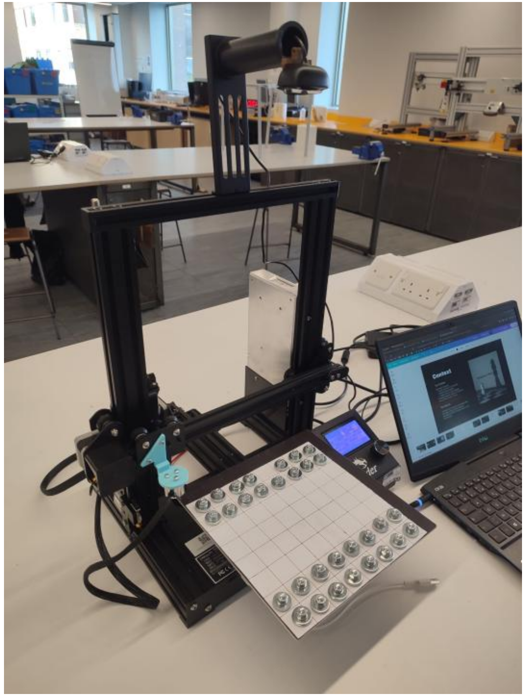
Team Coordinator

Stakeholder communication
Budget
Scheduling

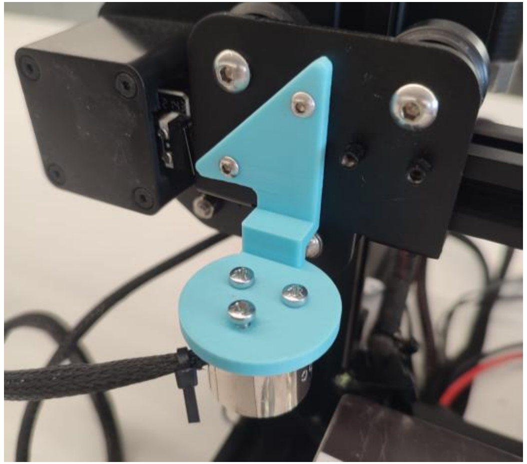


Mechanical Design

🖶 Creality Ender 3 Pro



🧲 Electromagnet End-Effector



RS PRO 150N DC electromagnet replaces hot-end. Custom PETG/PLA toolhead mount 3D-printed in-house.
Z-axis used purely for pick & place height.

♟ Custom 3D-Printed Pieces

Geometry optimised for consistent electromagnet engagement and centre-of-gravity stability.

Kinematics

Why IK is Closed-Form for a Cartesian Robot

Three independent prismatic joints: each joint controls exactly one Cartesian axis. The Jacobian is the **3×3 identity matrix**, so IK reduces to direct assignment with a unique, singularity-free solution.

FK: $p = J \cdot q$
$(X,Y,Z) = I_3 \cdot (q_1,q_2,q_3)$

invert →

IK: $q = J^{-1} \cdot p$
$J^{-1} = I_3 \Rightarrow q = p$

$$q_1 = X = 26 + (f - 1) \times 27 \text{ mm}$$
$$q_2 = Y = 38 + (r - 1) \times 27 \text{ mm}$$
$$q_3 = Z \in \{ 18 \text{ mm (pick)}, 40 \text{ mm (transit)} \}$$

⇒ **No singularities & Unique solution**

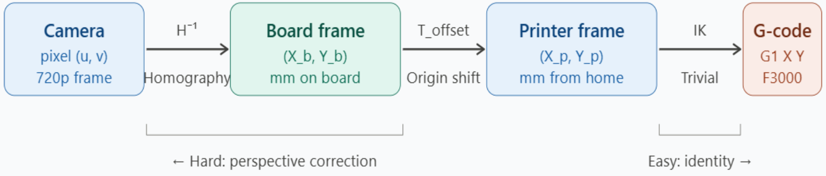
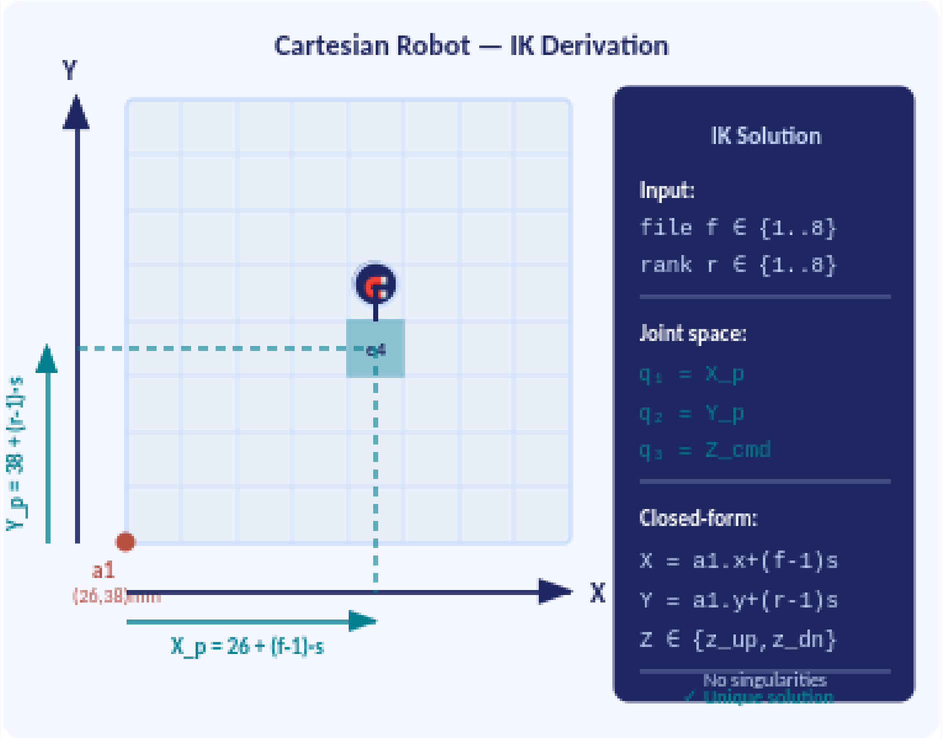


Fig 1: Coordinate transform chain — homography is the bottleneck, not IK



Computer Vision

Image Capture

LifeCam HD-3000 (720p/30fps) overhead bracket.

Perspective Correction

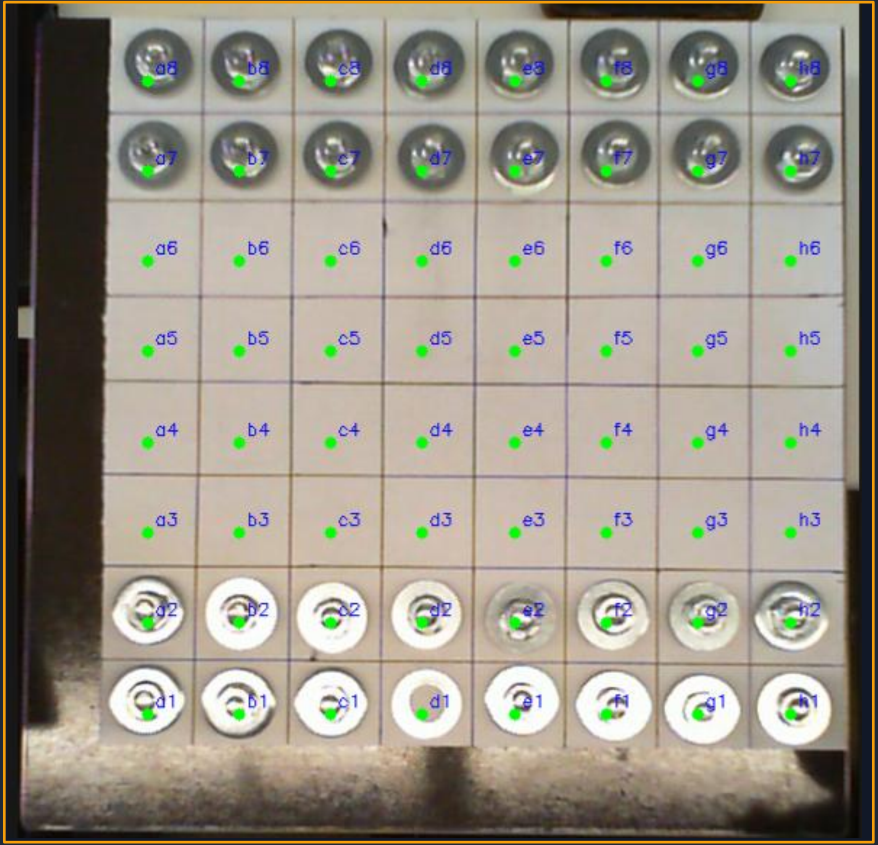
Homography → top-down rectified 8×8 view.

Grid Segmentation

64 equal-area regions from calibration JSON.

Move Detection

Frame diff → vacated & newly occupied squares.



OpenCV Grid Overlay

Motion Control & Chess Engine

Sunfish Chess Engine

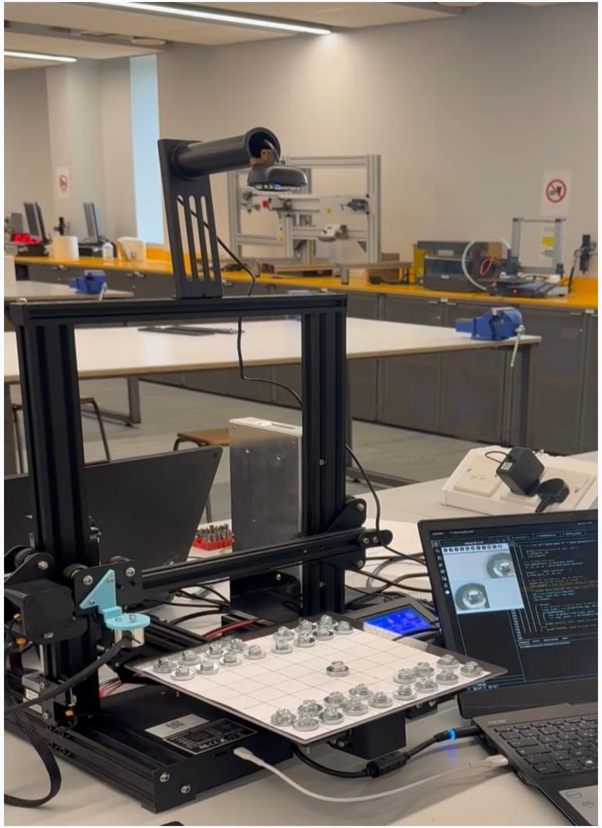
Compact Python engine: minimax search with alpha-beta pruning. Board state passed as string; returns best move in algebraic notation. Stateless: full game history passed each call.

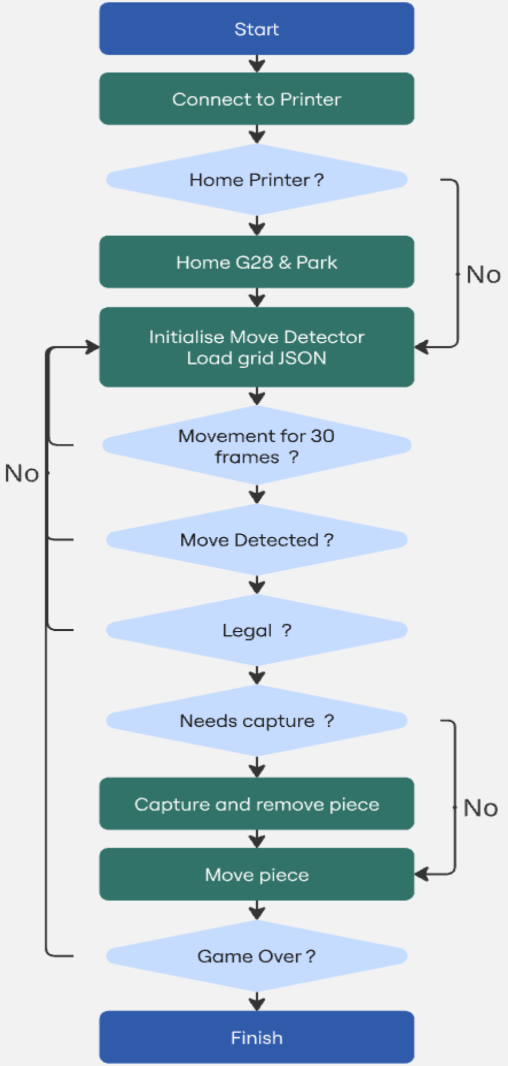
OctoPrint REST API

G-code streamed over HTTP to OctoPrint running locally. Decouples chess logic from printer firmware; no firmware modification needed. Conservative feed rates prevent piece displacement from inertia.

Captured Piece Logic

Captured pieces are relocated to a designated off-board holding area before the main piece moves, preventing physical collision on the board.





Bill of Materials & Cost

£0/50

Component	Specification	Market Value	Source / Actual Cost
Creality Ender 3 Pro	220×220×250mm Cartesian printer	~£180	GRID Lab (£0)
RS PRO Electromagnet	150N, 24V DC, 25mm dia., IP20	~£25	JN Mech. Workshop (£0)
Microsoft LifeCam HD	720p / 30fps overhead USB camera	~£30	Donated by lab (£0)
M3/M4 Screw Kit	Assorted hex bolts & washers	~£8	Lab stock (£0)
PLA Filament (×2 cols.)	Electromagnet mount + 32 chess pieces 1.75mm ~125g used	~£2	Lab printer stock (£0)
Laptop (control PC)	Python / OctoPrint / OpenCV host	—	Personal equipment (£0)

TOTAL ACTUAL COST: £0 out of £245

Conclusions

We successfully converted a Creality Ender 3 into a fully autonomous chess-playing robot, integrating computer vision, AI move generation, and electromagnetic actuation altogether for £0.

✅ All Objectives Met

- Full end-to-end autonomous gameplay demonstrated
- Human move detected via camera, zero manual input
- Detection success: 80%
  Pick-up success: ~95%
- Low-speed, constrained movements of robot
  Average move speed of ~15 seconds

⚙ Engineering Achieved

- Cartesian kinematics repurposed for chess workspace
- OpenCV homography:
  Pixel to G-code coordinate mapping
- Sunfish chess engine integrated into live loop
- OctoPrint used as firmware abstraction layer

💡 Key Learning

- Hard-coded grid calibration makes it easier to avoid lighting variation
- Lighting is the #1 enemy of vision-based pick-and-place
- Lead Screw is stable but slow
  Being slow it is safer for crushing incidents
- £0 spend proves DFMA
  + resource-reuse principles work

Future Work

Dynamic Difficulty

For the moment: sunfish only does the best moves

Add the option to choose Elo-rated engine levels.

Use the display to show information and handle settings

Voice Announcements

Microphone integration for spoken move announcements. This could help disabled people play chess with a physical board.

This could also allow two players to play against each other using only their voices.

Better piece handling

Have the robot store to eaten pieces on the side and add automatic piece repositioning.

Enable players to save their configuration to continue later.

Offer chess problems.

Video Demonstration

Calibration

System inquires whether printer needs to be homed (Y/N).
"Yes" input initiations system calibration of chess board grid.

Motion Capture and Cartesian Movement

- Machine vision captures human movement of opponent chess piece.
- Sunfish AI calculates best response movement.
- OctoPrint API converts movement to G-Code.
- Creality Ender 3 Pro translates G-code to corresponding movement.

Thank You

Questions?

Autonomous Chess Robot · B5#RO · 2025/2026

Arne Jacobs · Sai Neha Narendra Babu · Daniel McGurk · Vivek Devarapalli · Christian Lazaro

Creality Ender 3 · OpenCV · Sunfish Chess Engine · OctoPrint · Python